# OWASP Top 10 2021

The 10 most common web application vulnerabilities

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## Definition

The Open Worldwide Application Security Project (OWASP) is a nonprofit foundation dedicated to improving software security. It operates under an "open community" model, which means that anyone can participate in and contribute to OWASP-related online chats, projects, and more. For everything from online tools and videos to forums and events, the OWASP ensures that its offerings remain free and easily accessible through its website.

The OWASP Top 10 provides rankings of—and remediation guidance for—the top 10 most critical web application security risks. Leveraging the extensive knowledge and experience of the OWASP's open community contributors, the report is based on a consensus among security experts from around the world. Risks are ranked according to the frequency of discovered security defects, the severity of the uncovered vulnerabilities, and the magnitude of their potential impacts. The purpose of the report is to offer developers and web application security professionals insight into the most prevalent security risks so that they may fold the report's findings and recommendations into their own security practices, thereby minimizing the presence of known risks in their applications.

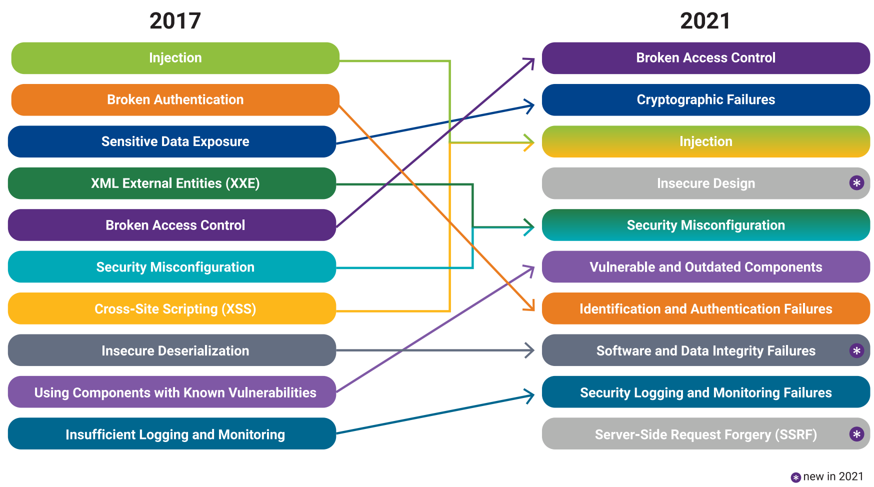

## How is the OWASP Top 10 list used and why is it important?

The OWASP has maintained its Top 10 list since 2003, updating it every two or three years in accordance with advancements and changes in the AppSec market. The list's importance lies in the actionable information it provides in serving as a checklist and internal web application development standard for many of the world's largest organizations.

Auditors often view an organization's failure to address the OWASP Top 10 as an indication that it may be falling short on other compliance standards. Conversely, integrating the Top 10 into the software development life cycle (SDLC) demonstrates an organization's overall commitment to industry best practices for secure development.

## What's new in the 2021 list?

For the 2021 list, the OWASP added three new categories, made four changes to naming and scoping, and did some consolidation.




# 1. Broken Access Control (A01:2021)

Previously number 5 on the list, broken access control—a weakness that allows an attacker to gain access to user accounts—moved to number 1 for 2021. The attacker in this context can function as a user or as an administrator in the system.

**Example:** An application allows a primary key to be changed, and when this key is changed to another user's record, that user's account can be viewed or modified.

**Solution:** An interactive application security testing (IAST) solution, such as Seeker®, can help you effortlessly detect cross-site request forgery or insecure storage of your sensitive data. It also pinpoints any bad or missing logic being used to handle JSON Web Tokens. Penetration testing can serve as a manual supplement to IAST activities, helping to detect unintended access controls. Changes in architecture and design may be warranted to create trust boundaries for data access.


## 2. Cryptographic Failures (A02:2021)

Previously in position number 3 and formerly known as sensitive data exposure, this entry was renamed as cryptographic failures to accurately portray it as a root cause, rather than a symptom. Cryptographic failures occur when important stored or transmitted data (such as a social security number) is compromised.

**Example:** A financial institution fails to adequately protect its sensitive data and becomes an easy target for credit card fraud and identity theft.

**Solution:** Seeker's checkers can scan for both inadequate encryption strength and weak or hardcoded cryptographic keys, and then identify any broken or risky cryptographic algorithms. The Black Duck® cryptography module surfaces the cryptographic methods used in open source software (OSS) so they can be further evaluated for strength. Both Coverity® static application security testing (SAST) and Black Duck software composition analysis (SCA) have checkers that can provide a "point in time" snapshot at the code and component levels. However, supplementing with IAST is critical for providing continuous monitoring and verification to ensure that sensitive data isn't leaked during integrated testing with other internal and external software components.


## 3. Injection (A03:2021)

Injection moves down from number 1 to number 3, and cross-site scripting is now considered part of this category. Essentially, a code injection occurs when invalid data is sent by an attacker into a web application in order to make the application do something it was not designed to do.

**Example:** An application uses untrusted data when constructing a vulnerable SQL call.

**Solution:** Including SAST and IAST tools in your continuous integration / continuous delivery (CI/CD) pipeline helps identify injection flaws both at the static code level and dynamically during application runtime testing. Modern application security testing (AST) tools such as Seeker can help secure the software application during the various test stages and check for a variety of injection attacks (in addition to SQL injections). For example, it can identify NoSQL injections, command injections, LDAP injections, template injections, and log injections. Seeker is the first tool to provide a new, dedicated checker designed to specifically detect Log4Shell vulnerabilities, determine how Log4J is configured, test how it actually behaves, and validate (or invalidate) those findings with its patented Active Verification engine.


## 4. Insecure Design (A04:2021)

Insecure design is a new category for 2021 that focuses on risks related to design flaws. As organizations continue to “shift left,” threat modeling, secure design patterns and principles, and reference architectures are not enough.

**Example:** A movie theater chain that allows group booking discounts requires a deposit for groups of more than 15 people. Attackers threat model this flow to see if they can book hundreds of seats across various theaters in the chain, thereby causing thousands of dollars in lost income.

**Solution:** Seeker IAST detects vulnerabilities and exposes all the inbound and outbound API, services, and function calls in highly complex web, cloud, and microservices-based applications. By providing a visual map of the data flow and endpoints involved, any weaknesses in the design of the app design are made clear, aiding in pen testing and threat modeling efforts.


## 5. Security Misconfiguration (A05:2021)

The former external entities category is now part of this risk category, which moves up from the number 6 spot. Security misconfigurations are design or configuration weaknesses that result from a configuration error or shortcoming.

**Example:** A default account and its original password are still enabled, making the system vulnerable to exploit.

**Solution:** Solutions like Coverity SAST include a checker that identifies the information exposure available through an error message. Dynamic tools like Seeker IAST can detect information disclosure and inappropriate HTTP header configurations during application runtime testing.


## 6. Vulnerable and Outdated Components (A06:2021)

This category moves up from number 9 and relates to components that pose both known and potential security risks, rather than just the former. Components with known vulnerabilities, such as CVEs, should be identified and patched, whereas stale or malicious components should be evaluated for viability and the risk they may introduce.

**Example:** Due to the volume of components used in development, a development team might not know or understand all the components used in their application, and some of those components might be out-of-date and therefore vulnerable to attack.

**Solution:** Software composition analysis (SCA) tools like Black Duck can be used alongside static analysis and IAST to identify and detect outdated and insecure components in an application. IAST and SCA work well together, providing insight into how vulnerable or outdated components are actually being used. Seeker IAST and Black Duck SCA together go beyond identifying a vulnerable component, uncovering details like whether that component is currently loaded by an application under test. Additionally, metrics such as developer activity, contributor reputation, and version history can give users an idea of the potential risk that a stale or malicious component may pose.

## 7. Identification and Authentication Failures (A07:2021)

Previously known as broken authentication, this entry has moved down from number 2 and now includes CWEs related to identification failures. Specifically, functions related to authentication and session management, when implemented incorrectly, allow attackers to compromise passwords, keywords, and sessions, which can lead to stolen user identity and more.

**Example:** A web application allows the use of weak or easy-to-guess passwords (i.e., “password1”).

**Solution:** Multifactor authentication can help reduce the risk of compromised accounts, and automated static analysis is highly useful in finding such flaws, while manual static analysis can add strength when evaluating custom authentication schemes. Coverity SAST includes a checker that specifically identifies broken authentication vulnerabilities. Seeker IAST can detect hardcoded passwords and credentials, as well improper authentication or missing critical steps in authentication.


## 8. Software and Data Integrity Failures (A08:2021)

This is a new category for 2021 that focuses on software updates, critical data, and CI/CD pipelines used without verifying integrity. Also now included in this entry, insecure deserialization is a deserialization flaw that allows an attacker to remotely execute code in the system.

**Example:** An application deserializes attacker-supplied hostile objects, opening itself to vulnerability.

**Solution:** Application security tools help detect deserialization flaws, and penetration testing can validate the problem. Seeker IAST can also check for unsafe deserialization and help detect insecure redirects or any tampering with token access algorithms.


## 9. Security Logging and Monitoring Failures (A09:2021)

Formerly known as insufficient logging and monitoring, this entry has moved up from number 10 and has been expanded to include more types of failures. Logging and monitoring are activities that should be performed on a website frequently—failure to do so leaves a site vulnerable to more severe compromising activities.

**Example:** Events that can be audited, like logins, failed logins, and other important activities, are not logged, leading to a vulnerable application.

**Solution:** After performing penetration testing, developers can study test logs to identify possible shortcomings and vulnerabilities. Coverity SAST and Seeker IAST can help identify unlogged security exceptions.


## 10. Server-Side Request Forgery (A10:2021)

A new category this year, a server-side request forgery (SSRF) can happen when a web application fetches a remote resource without validating the user-supplied URL. This allows an attacker to make the application send a crafted request to an unexpected destination, even when the system is protected by a firewall, VPN, or additional network access control list. The severity and incidence of SSRF attacks are increasing due to cloud services and the increased complexity of architectures.

**Example:** If a network architecture is unsegmented, attackers can use connection results or elapsed time to connect or reject SSRF payload connections to map out internal networks and determine if ports are open or closed on internal servers.

**Solution:** Seeker is one of the modern AST tools that can track, monitor, and detect SSRF without the need for additional scanning and triaging. Due to its advanced instrumentation and agent-based technology, Seeker can pick up any potential exploits from SSRF as well.


## How can Black Duck help?

Most businesses use a multitude of application security tools to help check off OWASP compliance requirements. While this is a good application security practice, it is not sufficient—organizations still face the challenge of aggregating, correlating, and normalizing the different findings from their various AST tools. This is where an application security posture management (ASPM) solution will improve process efficiency and team productivity.

Having an ASPM solution can aid in proactively tracking and addressing violations of OWASP Top 10 standards. ASPM solutions like Software Risk Manager can contextualize high-impact security activities based on their assessment of application risk and compliance violations.

These solutions offer a frictionless means to visualize and implement OWASP standards early because they integrate with developer frameworks and tools that support continuous testing, tracking, and management of security activities and findings. Software Risk Manager, for example, can centrally consume results from all AST tools (SAST, DAST, SCA, open source, and commercial), correlate these findings, consolidate them by type, and then enable users to view which findings constitute violations of OWASP standards through the built-in compliance reporting capability.

Additional testing can determine the type of testing required and the business criticality of the application to be tested. While AST tools offer valuable information to address individual OWASP standards, an ASPM approach can help facilitate and orchestrate repeatable software quality control and operations across all AST issues.

# Resources to manage your enterprise AppSec risk

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